NEET-SS / FET / INI-SS HIGH-YIELD NOTES

Arthroplasty

1. THA approaches and complications
2. TKA balancing and alignment

THA approaches and complications

1. Indications and pre-operative planning

- **Indications:** end-stage hip arthritis (primary OA, inflammatory arthritis, AVN, post-traumatic, DDH, Perthes/SCFE sequelae), **displaced femoral neck fractures in active elderly**, failed fixation, tumours.
- **Contraindications:** active infection (joint or systemic), neuropathic (Charcot) joint (relative), severe abductor deficiency (relative: use constrained/dual mobility), medically unfit.
- **Templating** (calibrated AP pelvis, magnification about 20%): **restore the centre of rotation, femoral offset and leg length**; cup at the **teardrop** (medial border) and inferior at the teardrop line; femoral component size and neck cut level; **offset** (reduced offset weakens the abductors and increases impingement/dislocation; excessive offset causes trochanteric pain).
- **Leg length:** measured to the **inter-teardrop line** from the lesser trochanters; **lengthening over about 2-4 cm** risks **sciatic nerve palsy** (peroneal division).

2. Surgical approaches

Approach	Interval / technique	Advantages	Risks
Posterior (Southern/Moore, Kocher-Langenbeck)	Split gluteus maximus; release **short external rotators (piriformis, gemelli, obturator internus) and posterior capsule**	**Commonest**, extensile, preserves abductors, good femoral exposure	**Higher dislocation (posterior)** (reduced by **capsule and rotator repair**, about 1% with repair vs 4% without), **sciatic nerve** injury
Direct lateral (Hardinge, transgluteal)	**Split gluteus medius** (anterior third to half) and vastus lateralis (no true internervous plane)	**Low dislocation**, good exposure	**Superior gluteal nerve** injury if the split goes **over about 5 cm above the greater trochanter tip** (abductor weakness, **Trendelenburg gait/limp**), heterotopic ossification
Anterolateral (Watson-Jones)	Between **tensor fasciae latae and gluteus medius** (both superior gluteal nerve: not a true internervous plane)	Low dislocation	Superior gluteal nerve (inferior branch to TFL), abductor damage

Approach	Interval / technique	Advantages	Risks
Direct anterior (Smith-Petersen/Hueter)	**Superficial: sartorius (femoral) and TFL (superior gluteal)**; **deep: rectus femoris (femoral) and gluteus medius (superior gluteal)**: true internervous plane	Muscle-sparing, **low dislocation**, supine position (fluoroscopy, leg length), possibly faster early recovery	**Lateral femoral cutaneous nerve (meralgia paraesthetica; up to 15-80% transient numbness)**, **femoral fractures** (calcar, greater trochanter) during broaching, wound problems in obese patients, learning curve, **ascending branch of LFCA** (ligate)
Trochanteric osteotomy (Charnley)	Greater trochanter osteotomised	Wide exposure (complex primaries, revision)	**Trochanteric non-union/escape**, wire problems

- **Long-term outcomes are similar** between approaches; choose by surgeon experience and patient factors.
- **Femoral neurovascular structures** lie anterior: anterior retractors placed too medially over the anterior acetabular rim can injure the **femoral nerve/artery**.

3. Acetabular component position

- **Lewinnek "safe zone": inclination (abduction) 40 ±10 degrees**, **anteversion 15 ±10 degrees** (radiographic). Many dislocations occur within this zone, so the **functional (spinopelvic) position** matters: a

stiff lumbar spine (fusion, degenerative) changes the safe zone (adjust anteversion, consider dual mobility).

- **Combined anteversion (cup + stem) about 25-45 degrees** (Ranawat test/Dorr).
- **Wallace/Hip-spine:** see spinopelvic notes.
- **Steep inclination (over 50 degrees)** = edge loading, increased wear, **squeaking (ceramic-on-ceramic)**, metal ion release (metal-on-metal), dislocation. **Excessive anteversion** = anterior dislocation; **retroversion** = posterior dislocation.
- **Acetabular screws (Wasielewski quadrant system):** a line from the ASIS through the centre of the acetabulum, and a perpendicular: **posterosuperior quadrant = safe** (screws up to about 25 mm+; sciatic nerve and superior gluteal vessels if too long), **posteroinferior = relatively safe** (sciatic, inferior gluteal, internal pudendal), **anterosuperior = "death zone" (external iliac artery and vein)**, **anteroinferior = obturator nerve and vessels**.

4. Fixation

- **Cemented femoral stems: taper-slip (polished, collarless: Exeter, CPT)** subside within the cement mantle creating radial compressive forces (excellent long-term survival), or **composite-beam (shape-closed, matt, collared: Charnley)** bonded to cement. **Third-generation cementing: pulsatile lavage, distal cement restrictor/plug, vacuum mixing (reduces porosity), retrograde filling with a gun, pressurisation, stem centralisers**, mantle **2 mm or more** (avoid thin mantle/defects: Barrack grading A-D).
- **Uncemented stems:** fixation by **bone ingrowth (porous coating, pore size 50-400 microns)** or **ongrowth (grit-blasted)**; **hydroxyapatite coating** enhances osseointegration; **micromotion under about 40-150 microns** required. **Stress shielding:** proximal bone loss with **stiff, large, extensively coated stems** (Engh); proximal femoral bone loss (Gruen zones 1 and 7).

- **Cups:** mostly uncemented (press-fit, **under-reaming 1-2 mm**, ± screws); cemented polyethylene cups in elderly/poor bone.
- **Age-related trends:** cemented stems reduce **periprosthetic fracture** risk in the elderly (registries: **uncemented stems have higher early periprosthetic fracture rates over about 75 years**).
- **Gruen zones (femur):** 7 zones on AP (1 greater trochanter, 4 tip, 7 calcar), 14 including lateral; **DeLee-Charnley zones (acetabulum)** I-III.

5. Complications

Dislocation

- **About 1-3% after primary THA**; the commonest cause of early revision in some registries.
- **Risk factors: posterior approach without repair**, **component malposition**, **small head size** (**larger heads (32-36 mm) increase jump distance and impingement-free arc**), **reduced offset/soft tissue tension**, abductor deficiency, **neuromuscular disease, cognitive impairment, alcohol**, **spinal fusion/stiffness**, **revision surgery**, female sex, **previous hip surgery**, **fracture as indication**.
- **Early dislocations** (within 3 months) with well-positioned components: **closed reduction** and precautions/brace (about 2/3 do not recur). **Recurrent or malpositioned:** revise: reposition components, **dual-mobility** cup, **larger head**, **constrained liner** (last resort, higher failure), **trochanteric advancement**, abductor reconstruction.
- **Dual-mobility bearings:** small head within a large mobile polyethylene liner inside a polished shell: large effective head, very low dislocation; **intraprosthetic dislocation** is a unique complication.

Periprosthetic fracture (Vancouver classification, femur)

Type	Location	Stem	Treatment
A_G / A_L	Greater / lesser trochanter	Stable	Non-operative unless displaced (A_G: fix if displaced over 2.5 cm/abductor function)
B1	Around/just below the stem	**Stable (well-fixed)**	**ORIF** (locking plate + cables, ± strut allografts)
B2	Around the stem	**Loose**, good bone stock	**Revision with a long stem** (distal fixation, e.g. tapered fluted modular) bypassing the fracture by 2 cortical diameters
B3	Around the stem	**Loose, poor bone stock**	**Revision with proximal femoral replacement** (elderly) or allograft-prosthesis composite (young)
C	Well below the stem tip	Stable	**ORIF** (overlap the stem to avoid stress risers)

- **Distinguishing B1 from B2 is critical**: **a significant proportion of "B1" are actually loose**: test stability intraoperatively; look for radiolucent lines, subsidence, cement mantle fracture.
- **Intraoperative fractures:** commoner with **uncemented press-fit stems**; calcar cracks: **cerclage cable**.
- **Unified Classification System (UCS)** applies to all periprosthetic fractures (see paediatric/trauma notes for type definitions: A apophyseal, B bed of implant (B1 stable, B2 loose, B3 poor bone), C clear of implant, D dividing two implants, E each of two bones, F facing an implant).
- **Acetabular periprosthetic fractures (Paprosky).**

Leg-length discrepancy

- Commonest cause of litigation; lengthening under 1 cm usually tolerated. **True LLD** (bony) vs **apparent/functional** (pelvic obliquity,

contractures). **Sciatic palsy risk with lengthening over about 2.5-4 cm.**

Nerve injury (about 0.5-3%)

- **Sciatic nerve (peroneal division)** commonest (posterior approach, lengthening, DDH/revision, retractor placement, cement); **femoral nerve** (anterior retractors), **superior gluteal nerve** (lateral approach), **lateral femoral cutaneous** (anterior approach), obturator (screws/cement).

Heterotopic ossification (HO)

- **Brooker classification: I** bone islands in soft tissue; **II** spurs with **over 1 cm** between opposing surfaces; **III** under 1 cm gap; **IV** apparent **bony ankylosis**.
- **Risk factors: ankylosing spondylitis, DISH, previous HO**, hypertrophic OA, Paget disease, **male sex**, **lateral/transtrochanteric approaches**, head injury.
- **Prophylaxis** (high-risk patients): **NSAIDs (indomethacin for about 6 weeks)** or **single-dose radiotherapy (about 7-8 Gy) within 24 hours before or 72 hours after surgery**. Excision of mature HO (normal alkaline phosphatase, bone scan quiescent, usually after 6-12 months) with prophylaxis.

Venous thromboembolism

- **DVT without prophylaxis up to about 40-60%**; fatal PE about 0.1-0.5%. **Prophylaxis: LMWH, rivaroxaban/apixaban/dabigatran (DOACs), or aspirin** (**CRISTAL trial** (2022): aspirin had more symptomatic VTE than enoxaparin after arthroplasty; **PEPPER trial**: aspirin, warfarin, rivaroxaban compared), plus **mechanical prophylaxis and early mobilisation**. Extended prophylaxis about **28-35 days after THA** (about 10-14 days after TKA).

Others

- **Infection (about 1%)**, see PJI notes. **Aseptic loosening** and **osteolysis** (wear particles: see bearing notes), **trunnionosis/taper corrosion** (head-neck junction; metal ions; ALTR), **squeaking** (CoC), **iliopsoas impingement** (cup prominent anteriorly; groin pain on resisted hip flexion; **iliopsoas tenotomy or cup revision**), **greater trochanteric pain syndrome**, **vascular injury**, **bone cement implantation syndrome** (hypotension, hypoxia, cardiac arrest during cementing, especially elderly with fractures), **medical complications**.

6. Special situations

- **DDH (Crowe classification by proximal migration of the head): I under 50%**, **II 50-75%**, **III 75-100%**, **IV over 100%** subluxation (relative to femoral head height). **Hartofilakidis** (dysplasia, low dislocation, high dislocation). Place the **cup at the true acetabulum** (anatomical centre), small cups, structural autograft/medialisation, **subtrochanteric shortening osteotomy** (Crowe IV) to prevent sciatic stretch; femur often **narrow and excessively anteverted** (modular or cemented stems to control version).
- **Protrusio acetabuli** (rheumatoid arthritis, Paget, Marfan, osteomalacia): **medial wall bone grafting**, lateralise the centre of rotation, cages/rings in severe cases.
- **Paget disease:** bleeding (consider pre-op bisphosphonates), deformity (osteotomy), hard sclerotic bone, HO.
- **Sickle cell, post-fixation, post-acetabular fracture, ankylosed hip** (abductor function uncertain), **neuromuscular hips**, **young patients** (bearing choice).

Exam pearl: **Posterior approach = commonest, higher dislocation without capsular repair; Hardinge = superior gluteal nerve 5 cm above the trochanter (Trendelenburg); direct anterior = lateral femoral cutaneous nerve and femoral fractures.**

Exam pearl: **Lewinnek: inclination 40 ±10, anteversion 15 ±10. Anterosuperior acetabular quadrant = external iliac vessels (death zone); posterosuperior = safe.** Larger heads and dual mobility reduce dislocation.

Exam pearl: **Vancouver B1 stable = ORIF; B2 loose = long-stem revision; B3 loose + poor bone = proximal femoral replacement/APC; C = ORIF. Brooker HO I-IV; prevent with indomethacin or radiotherapy (7-8 Gy) in high-risk patients.**

TKA balancing and alignment

1. Alignment concepts

- **Mechanical axis of the lower limb:** centre of the femoral head to the centre of the ankle; normally passes **just medial to the centre of the knee** (about 0-3 degrees varus overall).
- **Anatomical vs mechanical femoral axis:** differ by about **5-7 degrees (valgus)**; the **distal femoral cut is made at about 5-7 degrees valgus to the anatomical axis** (perpendicular to the mechanical axis).
- **Joint line:** the native tibial plateau is in **about 3 degrees varus** relative to the tibial mechanical axis; the distal femur in about 3 degrees valgus relative to the femoral mechanical axis; the **joint line is roughly parallel to the floor in stance** (with feet together).
- **Alignment philosophies:**

Philosophy	Principle	Notes
Mechanical alignment (MA)	Femoral and tibial cuts **perpendicular to their mechanical axes** (neutral HKA, joint line horizontal); balance by **soft-tissue releases**	Traditional gold standard; **neutral axis ±3 degrees** associated with better survival (debated with modern implants)
Anatomical alignment (Hungerford)	Joint line at 3 degrees varus, neutral limb	Historical
Kinematic alignment (KA)	**Resurface the pre-arthritic joint** (restore native joint lines and HKA), **minimal releases**	Implants positioned within the patient's native alignment; outlier limb alignments; satisfaction similar or slightly better in some trials

Philosophy	Principle	Notes
Restricted KA / inverse kinematic / functional alignment	KA within safe limits; robotic gap-based adjustments	**CPAK classification** (Coronal Plane Alignment of the Knee: arithmetic HKA and joint line obliquity; 9 phenotypes) describes constitutional alignment

- **Robotic/navigation** TKA reduces alignment outliers; clinical superiority not yet proven.

2. Bone cuts and component rotation

- **Tibial cut:** perpendicular to the tibial mechanical axis in the coronal plane (MA), **posterior slope about 3-7 degrees** (CR knees more; PS knees less, as excess slope can cause instability in flexion/post impingement). Resect about **8-10 mm from the unaffected (higher) side** (measured resection).
- **Distal femoral cut:** about 5-7 degrees valgus to the anatomical axis (intramedullary guide entry point just medial and anterior to the PCL origin); resection equal to the implant thickness (about 9-10 mm).
- **Femoral component rotation (critical for patellar tracking and flexion gap balance):**
  - **Transepicondylar axis (TEA)** (lateral epicondyle to the **medial epicondylar sulcus**: surgical TEA): **most reliable** rotational landmark.
  - **Whiteside line (AP axis):** from the deepest part of the trochlear groove to the centre of the intercondylar notch; femoral component **perpendicular** to it (unreliable in trochlear dysplasia).
  - **Posterior condylar axis (PCA):** femoral component **externally rotated about 3 degrees** relative to it (because the medial posterior condyle is larger); **beware a hypoplastic lateral**

condyle in valgus knees (would internally rotate the component: use TEA/Whiteside).

- **Gap balancing technique:** rotate the femoral component so the **flexion gap is rectangular** after the tibial cut with balanced ligaments.

- **Tibial component rotation:** aligned to the **medial third of the tibial tubercle** (Akagi line: PCL attachment to the medial border of the patellar tendon at its tibial attachment).
- **Internal rotation of femoral or tibial components** (combined) causes **patellar maltracking, lateral subluxation, anterior knee pain, patellar complications and stiffness** (Berger: combined internal rotation 1-4 degrees = tilt; 3-8 = subluxation; 7-17 = dislocation/failure). Assess with **CT**.
- **Anterior femoral notching** (anterior cortex): increases **supracondylar periprosthetic fracture** risk (about 30-40% reduction in torsional strength).
- **Femoral sizing: anterior referencing** (cuts the posterior condyles by the measured amount; downsizing changes the flexion gap: may loosen flexion) vs **posterior referencing** (restores posterior condylar offset; risk of anterior notching or overstuffing the patellofemoral joint).

3. Gap balancing

- **Extension gap** (distal femoral cut + tibial cut) and **flexion gap** (posterior femoral cut + tibial cut) must be **equal and rectangular**.
- **The tibial cut affects both gaps; the distal femoral cut affects only the extension gap; the posterior femoral cuts/femoral size affect only the flexion gap.**

Problem	Flexion gap	Extension gap	Solution
Tight in both	Tight	Tight	**Cut more tibia**

Problem	Flexion gap	Extension gap	Solution
Loose in both	Loose	Loose	**Thicker polyethylene insert** (or augments)
Tight in extension, OK in flexion	Balanced	Tight	**Release posterior capsule / remove posterior osteophytes**, then **resect more distal femur** (about 2 mm)
Loose in extension, OK in flexion	Balanced	Loose	**Distal femoral augmentation** (add distal femur), thicker insert + smaller/posterior cut (combined)
Tight in flexion, OK in extension	Tight	Balanced	**Downsize the femoral component** (anterior referencing) or **increase tibial slope**; release PCL (CR)
Loose in flexion, OK in extension	Loose	Balanced	**Upsize femur** (posterior augments) or **thicker insert + more distal femoral resection**

- **Joint line:** keep within about **8 mm** of native: **elevation of the joint line** (excess distal femoral resection + thick insert) causes **mid-flexion instability, patella baja (pseudo), patellofemoral problems**.

4. Coronal deformity correction (soft-tissue releases)

Varus knee (commonest)

- **Tight medial structures:** remove **medial osteophytes first** (tibial and femoral), then release **deep MCL (meniscotibial)**, **posteromedial capsule/semimembranosus** (tight in **extension**), **superficial MCL** (subperiosteal from tibia or **pie-crusting**; the **anterior sMCL is tight in flexion**, posterior in extension), **pes anserinus** (rarely). **PCL** release (in CR knees with tight flexion).

- Also: **medial tibial reduction osteotomy** (downsize/lateralise the tibial tray and remove medial bone).
- **Fixed flexion deformity:** remove posterior osteophytes, **posterior capsular release**, gastrocnemius recession; as a last resort extra distal femoral resection.

Valgus knee

- **Tight lateral structures: iliotibial band** (tight in **extension**), **popliteus** (tight in **flexion**), **lateral collateral ligament** (tight in **flexion and extension**), **posterolateral capsule** (extension), lateral head of gastrocnemius, biceps femoris.
- **Order (Ranawat inside-out pie-crusting, Whiteside):** posterolateral capsule, ITB, then popliteus/LCL as needed.
- **Hypoplastic lateral femoral condyle**: do not use the PCA alone for rotation.
- **Peroneal nerve palsy** risk when correcting **severe valgus with flexion contracture** (stretch): **flex the knee post-operatively**, loosen dressings; explore if no recovery.
- **Ranawat valgus types:** I (correctable, medial soft tissue intact), II (fixed, medial attenuated: may need constraint), III (severe osseous deformity with incompetent medial structures: constrained implant).

5. Constraint and design choices

Design	Mechanism	Indications
Cruciate-retaining (CR)	PCL retained; flatter or ultra-congruent inserts	Intact, functional PCL; less bone resection; **paradoxical anterior femoral sliding**

Design	Mechanism	Indications
Posterior-stabilised (PS)	**Cam-post** substitutes the PCL (**femoral rollback**)	**Absent/incompetent PCL**, **patellectomy**, inflammatory arthritis, large deformity, previous HTO (debated); risks: **post wear/fracture, patellar clunk syndrome** (fibrous nodule at the superior pole catching the box), cam jump (flexion instability)
Medial pivot / ultra-congruent	Highly conforming medial compartment	Mimics native kinematics
Varus-valgus constrained (CCK, non-linked)	Tall, wide post in a deep box	**MCL/LCL incompetence**, severe valgus, flexion-extension mismatch, revisions
Rotating-hinge (linked)	Hinged femur-tibia linkage with rotation	**Global ligamentous deficiency**, severe bone loss, distal femoral replacement (tumour/fracture), hyperextension instability, **neuromuscular recurvatum** (polio)

- **Use the least constraint needed** (more constraint = more stress transfer to fixation interfaces = loosening).
- **Mobile-bearing vs fixed-bearing**: no proven difference in survival; spinout risk with mobile bearings.
- **Patellar resurfacing:** reduces **reoperation for anterior knee pain** (meta-analyses, KAT trial: little functional difference); inflammatory arthritis favours resurfacing. **Patellar thickness should be restored** (avoid overstuffing). **Patellar complications:** maltracking (component internal rotation, lateral placement of femoral/tibial components, medial placement of patellar button is preferred),

fracture (over-resection, avascular after lateral release), **avascular necrosis**, **patellar clunk**, loosening, tendon rupture.

- **Patellar tracking: "no thumb" test**; lateral retinacular release if needed (protect the **superior lateral genicular artery**).

6. Complications of TKA (non-infective)

- **Stiffness/arthrofibrosis** (flexion under about 90 degrees at 6-12 weeks): **manipulation under anaesthesia (within about 6-12 weeks)**, arthrolysis, revision.
- **Instability: flexion instability** (flexion gap larger than extension; recurrent effusions, PS cam jump), **mid-flexion instability**, **genu recurvatum**, global instability: revise with balanced gaps/increased constraint.
- **Aseptic loosening/osteolysis**, **polyethylene wear**.
- **Periprosthetic fractures: supracondylar femur (Lewis-Rorabeck I undisplaced, II displaced + stable prosthesis (ORIF: locking plate or retrograde nail if open box), III loose prosthesis (revision/DFR))**; **tibial (Felix)**; **patella (Goldberg/Ortiguera)**.
- **Peroneal nerve palsy** (valgus, flexion contracture, epidural anaesthesia), **vascular injury** (popliteal artery behind the posterior capsule: **posterolateral corner, about 9 mm from the posterior tibia in flexion**).
- **Wound complications: use the most lateral previous incision** (lateral skin flap less perfused: blood supply from the medial side), skin bridges at least about 5-6 cm.
- **Extensor mechanism disruption:** patellar tendon rupture (**worst; augment with allograft or synthetic mesh, medial gastrocnemius flap**), quadriceps rupture.
- **VTE**, **persistent pain** (about 10-20% dissatisfied after TKA: highest predictors are pre-operative expectations, pain catastrophising, mild radiographic OA).

- **Unexplained painful TKA** work-up: **exclude infection first (ESR/CRP, aspiration)**, then instability, malrotation (CT), loosening, patellar issues, referred pain (hip, spine), CRPS, neuroma.

Exam pearl: **Tibial cut affects both gaps; distal femoral cut affects extension only; posterior femoral cut/femoral size affects flexion only. Tight both = cut more tibia; loose both = thicker insert.**

Exam pearl: **Femoral rotation: TEA most reliable; 3 degrees external to the posterior condylar axis; Whiteside line perpendicular. Internal rotation of components = patellar maltracking. Tibial rotation: medial third of the tubercle.**

Exam pearl: **Varus: medial osteophytes, deep MCL, posteromedial capsule/semimembranosus, then superficial MCL. Valgus: ITB (extension), popliteus (flexion), LCL (both); peroneal palsy risk with valgus + flexion contracture.** Least constraint necessary; rotating hinge for global ligament loss.